# Auditoría, Calidad y Fiabilidad Informáticas

*(IT Auditing, Quality, and Reliability)*

Universidad Complutense de Madrid  Academic Year 2026/27

Practical 1: Introduction to specification languages

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**Exercise 1** Specify in Maude a functional module `BST` for binary search trees. The module will have:

(a) Constructors for the empty search tree and for nodes with a key, a value, a left child, and a right child. Both keys and values will be natural numbers.

(b) A function `merge`, which takes two natural numbers and returns another natural number. This function will be used to combine two values when inserting a key that was already present, and it can behave however you like: for example, it can add the two values.

(c) An insertion function. This function uses `merge` to combine two values when the key being inserted already existed.

(d) A partial lookup function that, given a binary search tree and a key (a natural number), returns the associated value, if there is one.

**Exercise 2** Test the module:

(a) Inserting values into different trees, including the case in which the key is already present.

(b) Looking up values, including the case in which the key is not present.